

7. Variational Autoencoder

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MTH345: Deep Generative Models: A Statistical Perspective



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Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

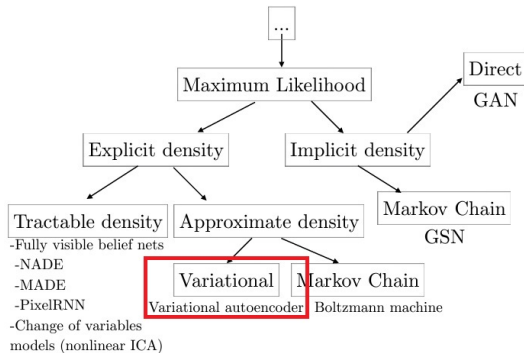
M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics



- ▶ In the previous chapter, we reviewed energy-based models. Both training and generation require running a Markov chain (with multi-step Gibbs sampling).
- ▶ In this lecture, we will review another generative model, *variational autoencoders* (VAEs). By applying variational inference, they do not require running Markov chains during either training or testing times.

The figure is edited from Goodfellow (2016).

Lecture Topics

- ▶ Recommended reference materials include:
 - ▶ Kingma and Welling (2014), "Auto-encoding variational Bayes," *Proceedings of the International Conference on Learning Representations (ICLR)*.
 - ▶ Kingma et al. (2019), "An Introduction to Variational Autoencoders," *Foundations and Trends® in Machine Learning*.
 - ▶ Goodfellow (2016), "Nips 2016 tutorial: Generative adversarial networks," *arXiv*.
- ▶ It is also encouraged to refer to *Chapter 7. Variational Autoencoders* in Kim (2026). Compared with that reference, this lecture note includes additional background on variational inference and covers roughly 30% of the VAE research addressed in the cited material.

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2. Variational Autoencoder
3. Multi-modal Variational Autoencoder
4. Temporal Variational Autoencoder

Motivation

- ▶ As a preliminary, we briefly review the framework of *variational inference* (Bishop, 2006).
- ▶ Consider a discrete mixture model with a latent variable $Z \in \{1, \dots, K\}$ and an observed variable X . The generative process is a Gaussian mixture model and is assumed to be known:¹

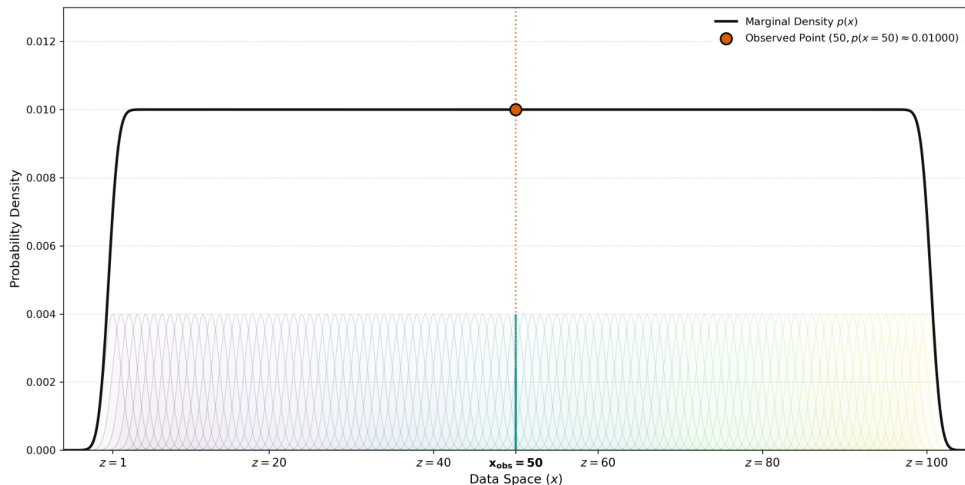
$$Z \sim \text{Multin}(1, (1/K, \dots, 1/K)^T) \iff p(z) = \frac{1}{K}, \quad \text{for } z = 1, \dots, K, \quad (1)$$

$$X | Z = z \sim \mathcal{N}(z, 1) \iff p(x | z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - z)^2}{2}\right). \quad (2)$$

- ▶ Evaluating the marginal likelihood $p(x)$ poses a severe computational challenge, and variational inference provides a prominent framework to circumvent this issue.

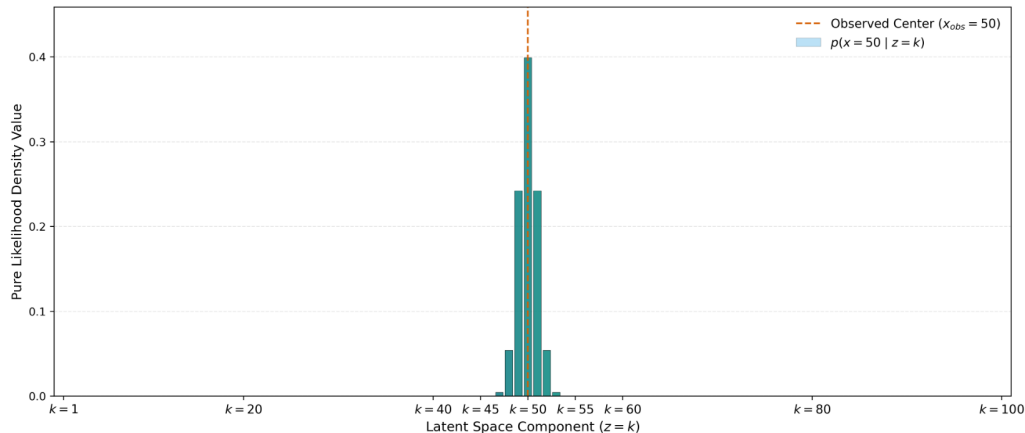
¹From now on, following the convention in variational inference, we use p to denote the PDF regardless of the data type.

Motivation



- The $p(x)$ can be expressed as $p(x) = \sum_{k=1}^K p(x | z)p(z) = \sum_{k=1}^K \phi(x - z)/K$.

Motivation



- Most of the values $p(x | z = k)$ are extremely close to zero, except within the narrow interval $k \in \{47, \dots, 53\}$.

Naive Monte Carlo

- ▶ When K is large or when Z is continuous, the analytical or numerical computation of the marginal likelihood $p(x)$ is often intractable.
- ▶ Here, we first consider an intuitive baseline approach.
- ▶ **Naive Monte Carlo**: Draw $z_{\text{naive}}^{(1)}, \dots, z_{\text{naive}}^{(M)} \stackrel{\text{i.i.d.}}{\sim} p(z)$ and calculate:

$$\hat{p}_{\text{naive}}(x) := \frac{1}{M} \sum_{m=1}^M p(x | z_{\text{naive}}^{(m)}). \quad (3)$$

- ▶ For example, when estimating $p(x = 50)$ with $M = 3$, if the drawn samples are $z_{\text{naive}}^{(1)} = 12$, $z_{\text{naive}}^{(2)} = 50$, and $z_{\text{naive}}^{(3)} = 84$, the empirical calculation yields:

$$\hat{p}_{\text{naive}}(x = 50) = \frac{0 + \frac{1}{\sqrt{2\pi}} + 0}{3} = \frac{1}{3\sqrt{2\pi}} \approx 0.13298. \quad (4)$$

Recall that the true marginal likelihood is

$$p(x = 50) = \frac{1}{100} \sum_{k=1}^{100} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(50-k)^2}{2}\right) \approx 0.01000.$$

Naive Monte Carlo

Naive Monte Carlo

- ▶ The empirical calculation results heavily depend on the samples falling to $\{47, \dots, 53\}$.
- ▶ Nevertheless, the Monte Carlo estimator is unbiased:

$$\mathbb{E}_{Z_{\text{naive}}^{(m)} \stackrel{\text{i.i.d.}}{\sim} p(z)} [\hat{p}_{\text{naive}}(x)] = \quad (5)$$

Naive Monte Carlo

- Its high variability can be expressed as:

$$\begin{aligned}\text{Var}_{Z_{\text{naive}}^{(m)} \stackrel{\text{i.i.d.}}{\sim} p(z)} [\hat{p}_{\text{naive}}(x)] &= \frac{1}{M} \text{Var}_{Z \sim p(z)} [p(x | Z)] \\ &= \frac{1}{M} \left(\mathbb{E}_{Z \sim p(z)} [p(x | Z)^2] - p(x)^2 \right).\end{aligned}\tag{6}$$

- By applying the CLT, as $M \rightarrow \infty$,

$$\begin{aligned}\sqrt{M} (\hat{p}_{\text{naive}}(x) - p(x)) &\xrightarrow{d} \mathcal{N} \left(0, \text{Var}_{Z \sim p(z)} [p(x | Z)] \right) \\ &= \mathcal{N} \left(0, \mathbb{E}_{Z \sim p(z)} [p(x | Z)^2] - p(x)^2 \right).\end{aligned}\tag{7}$$

Importance Sampling

- ▶ One main source of the high variability of the Naive Monte Carlo approach is that the prior $p(z = k)$ does not reflect the relative contribution of $p(x | z = k)$ to the marginal likelihood $p(x)$.
- ▶ Importance Sampling: Draw $z_{\text{is}}^{(1)}, \dots, z_{\text{is}}^{(M)} \stackrel{\text{i.i.d.}}{\sim} q(z)$ and calculate:

$$\hat{p}_{\text{is}}(x) := \sum_{m=1}^M p(x | z_{\text{is}}^{(m)}) \frac{p(z_{\text{is}}^{(m)})}{Mq(z_{\text{is}}^{(m)})}. \quad (8)$$

The $q(z)$ is referred to as the *proposal distribution*.

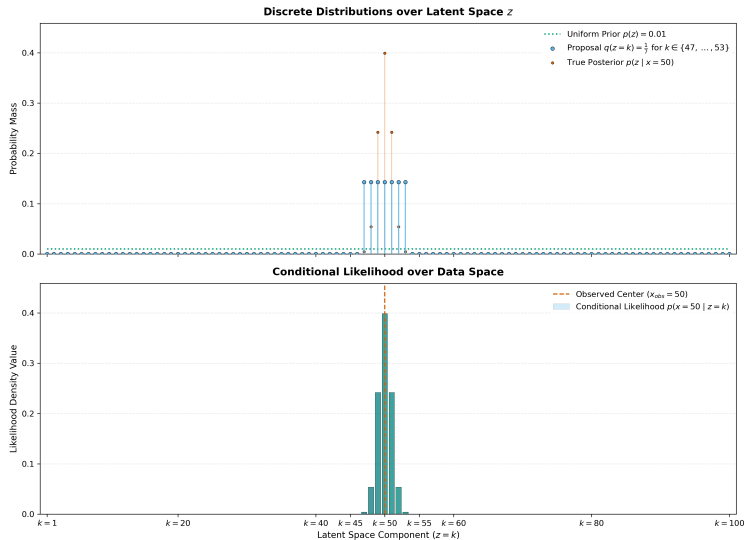
- ▶ By appropriately choosing $q(z)$ to yield high probability values where the joint density $p(x, z) = p(x | z)p(z)$ is significant, we can expect to reduce the variability of the estimator.

Importance Sampling

- ▶ The importance sampling estimator is unbiased:

$$\mathbb{E}_{Z_{\text{is}}^{(m)} \sim p(z)} [\hat{p}_{\text{is}}(x)] = \quad (9)$$

Importance Sampling



Importance Sampling

Importance Sampling

- ▶ The variance of the importance sampling estimator can be expressed as:

$$\begin{aligned}\text{Var}_{Z_{\text{is}}^{(m)} \text{i.i.d. } q(z)} [\hat{p}_{\text{is}}(x)] &= \frac{1}{M} \text{Var}_{Z \sim q(z)} \left[\frac{p(Z, x)}{q(Z)} \right] \\ &= \frac{1}{M} \left(\mathbb{E}_{Z \sim q(z)} \left[\left(\frac{p(Z, x)}{q(Z)} \right)^2 \right] - p(x)^2 \right).\end{aligned}\tag{10}$$

- ▶ By applying the CLT, we have:

$$\begin{aligned}\sqrt{M} (\hat{p}_{\text{is}}(x) - p(x)) &\xrightarrow{d} \mathcal{N} \left(0, \text{Var}_{Z \sim q(z)} \left[\frac{p(Z, x)}{q(Z)} \right] \right) \\ &= \mathcal{N} \left(0, \mathbb{E}_{Z \sim q(z)} \left[\left(\frac{p(Z, x)}{q(Z)} \right)^2 \right] - p(x)^2 \right).\end{aligned}\tag{11}$$

Importance Sampling

- Recall that:

$$\text{Var}_{Z_{\text{naive}}^{(m)} \stackrel{\text{i.i.d.}}{\sim} p(z)} [\hat{p}_{\text{naive}}(x)] = \frac{1}{M} \text{Var}_{Z \sim p(z)} [p(x | Z)]. \quad (12)$$

- We derived that:

$$\text{Var}_{Z_{\text{is}}^{(m)} \stackrel{\text{i.i.d.}}{\sim} q(z)} [\hat{p}_{\text{is}}(x)] = \frac{1}{M} \text{Var}_{Z \sim q(z)} \left[\frac{p(Z, x)}{q(Z)} \right]. \quad (13)$$

- By Jensen's inequality, we have:

$$\text{Var}_{Z_{\text{is}}^{(m)} \stackrel{\text{i.i.d.}}{\sim} q(z)} [\hat{p}_{\text{is}}(x)] = \frac{1}{M} \left(\mathbb{E}_{Z \sim q(z)} \left[\left(\frac{p(Z, x)}{q(Z)} \right)^2 \right] - p(x)^2 \right) \geq 0. \quad (14)$$

The equality holds if and only if $q^*(z) = p(z | x)$.

Variational Inference

- ▶ The ideal choice for $q(z)$, the true posterior $q^*(z) = p(z | x)$, is often computationally intractable.
- ▶ **Variational Inference**: We can reformulate the intractable inference problem into a deterministic optimization framework:

$$q_{\phi^*} = \arg \min_{q_{\phi}(z) \in \mathcal{Q}} \mathcal{D}_{\text{KL}}(q_{\phi}(z) \parallel p(z | x)), \quad (15)$$

where $\mathcal{Q}$ denotes a tractable family of distributions, a paradigm also known as *Variational Bayes*.

- ▶ Since q_{ϕ^*} depends on the observed data x , we map each observation to its corresponding data-dependent proposal:

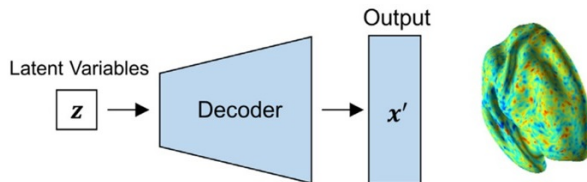
$$x \mapsto q_{\phi^*}(z | x) := \arg \min_{q_{\phi}(z|x) \in \mathcal{Q}} \mathcal{D}_{\text{KL}}(q_{\phi}(z | x) \parallel p(z | x)). \quad (16)$$

- ▶ The *Variational Autoencoder* (VAE) operationalizes this generalized encoding process by parameterizing the mapping via a neural network.

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Generation with Nonlinear Decoder



- ▶ How about generating data using a decoder? That is, the density is modeled as

$$p_{\theta}(\mathbf{z}, \mathbf{x}) := p(\mathbf{z})p_{\theta}(\mathbf{x} | \mathbf{z}), \quad (17)$$

where $p(\mathbf{z})$ be prior (e.g., Gaussian) and $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_{\theta}(\mathbf{z}), \Sigma_{\theta}(\mathbf{z}))$ with $\boldsymbol{\mu}_{\theta}(\cdot)$ and $\Sigma_{\theta}(\cdot)$ being neural networks parameterized by θ .

- ▶ This approach extends the decoder ($\boldsymbol{\mu}_{\theta}$) to a probabilistic model. When we sample $\mathbf{Z}$ multiple times from $p(\mathbf{z})$ and sample $\mathbf{X}$ from this (conditional) Gaussian, the distribution of generated $\mathbf{X}$ would be: $p_{\theta}(\mathbf{x}) = \int p(\mathbf{z})p_{\theta}(\mathbf{x}|\mathbf{z})d\mathbf{z}$.

Images are edited from Kim et al. (2021).

Generation with Nonlinear Decoder

- ▶ Due to the flexibility of nonlinear decoder networks, finding the MLE becomes highly challenging:

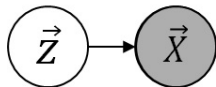
$$\theta^* = \arg \max_{\theta} \sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i). \quad (18)$$

- ▶ Note that the marginal likelihood $p_{\theta}(\mathbf{x}_i) := \int p(\mathbf{z})p_{\theta}(\mathbf{x}_i | \mathbf{z})d\mathbf{z}$ requires integrating highly nonlinear functions, e.g.,

$$\begin{aligned} & p(\mathbf{z})p_{\theta}(\mathbf{x}_i | \mathbf{z}) \\ & \propto \exp(-\mathbf{z}^T \mathbf{z} / 2) \cdot |\Sigma_{\theta}(\mathbf{z})|^{-1/2} \exp(-(\mathbf{x}_i - \boldsymbol{\mu}_{\theta}(\mathbf{z}))^T \Sigma_{\theta}(\mathbf{z})^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_{\theta}(\mathbf{z})) / 2). \end{aligned} \quad (19)$$

- ▶ Furthermore, the inference with $p_{\theta}(\mathbf{z} | \mathbf{x}) := p_{\theta}(\mathbf{z}, \mathbf{x}) / p_{\theta}(\mathbf{x})$ is also less straightforward.

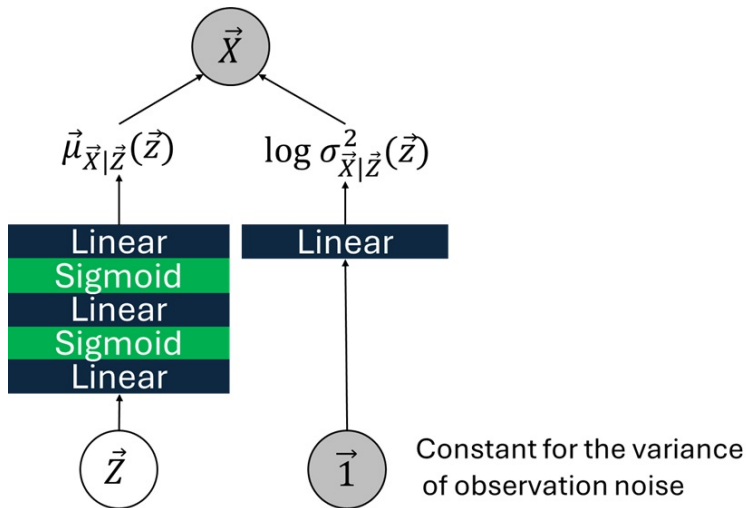
Variational Autoencoder



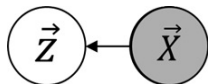
- ▶ Variational Autoencoders (VAEs, Kingma and Welling, 2014) utilize variational inference to overcome the aforementioned challenges.
- ▶ They model the data generation process using decoder networks with a diagonal covariance $\Sigma_\theta := \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$:

$$\begin{aligned}
 p_\theta(\mathbf{z}, \mathbf{x}) &= \left(\prod_{j=1}^r p(z_j) \right) \left(\prod_{j=1}^d p_\theta(x_j | \mathbf{z}) \right) \\
 &= \prod_{j=1}^r \left(\frac{1}{\sqrt{2\pi}} \exp \left(-\frac{z_j^2}{2} \right) \right) \prod_{j=1}^d \left(\frac{1}{\sqrt{2\pi\sigma_j^2}} \exp \left(-\frac{(x_j - (\mu_\theta(\mathbf{z}))_j)^2}{2\sigma_j^2} \right) \right).
 \end{aligned}
 \tag{20}$$

Variational Autoencoder



Variational Autoencoder



- ▶ VAEs introduce an encoder, or *inference network*, $q_{\phi}(\mathbf{z}|\mathbf{x})$ to approximate $p_{\theta}(\mathbf{z}|\mathbf{x})$.²
- ▶ By applying the variational inference, we have:

$$\log p_{\theta}(\mathbf{x}) = \log \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\frac{p_{\theta}(\mathbf{z}, \mathbf{x})}{q_{\phi}(\mathbf{z} | \mathbf{x})} \right]. \quad (21)$$

²VAEs use $q_{\phi}(\mathbf{z}|\mathbf{x})$, a single function (of $\mathbf{x}$), to approximate the posterior distribution for each new data point, referred to as *amortized variational inference*.

Variational Autoencoder

- ▶ The process of finding the optimal proposal q_{ϕ^*} can be interpreted as maximizing a lower bound of the evidence $\log p_{\theta}(\mathbf{x})$:

$$\log p_{\theta}(\mathbf{x}) \geq \log p_{\theta}(\mathbf{x}) - \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z} | \mathbf{x}) \parallel p_{\theta}(\mathbf{z} | \mathbf{x})). \quad (22)$$

- ▶ Here, the RHS

$$\text{ELBO}(\theta, \phi; \mathbf{x}) := \log p_{\theta}(\mathbf{x}) - \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z} | \mathbf{x}) \parallel p_{\theta}(\mathbf{z} | \mathbf{x})), \quad (23)$$

is referred to as the *evidence lower bound (ELBO)*, which exhibits a computationally tractable form:

$$\text{ELBO}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{\mathbf{Z} \sim q_{\phi}(\mathbf{z} | \mathbf{x})} [\log p_{\theta}(\mathbf{x} | \mathbf{Z})] - \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z} | \mathbf{x}) \parallel p(\mathbf{z})), \quad (24)$$

Variational Autoencoder

► $\text{ELBO}(\theta, \phi; \mathbf{x}) := \log p_{\theta}(\mathbf{x}) - \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z} \mid \mathbf{x}) \parallel p_{\theta}(\mathbf{z} \mid \mathbf{x})).$

Variational Autoencoder

- ▶ In the ELBO,

$$\text{ELBO}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{\mathbf{Z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x} | \mathbf{Z})] - \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z} | \mathbf{x}) \parallel p(\mathbf{z})), \quad (25)$$

the first term on the RHS represents the *reconstruction probability*, and maximizing it can be viewed as training a probabilistic autoencoder.

- ▶ For example, when $q_{\phi}(\mathbf{z} | \mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_{\phi}(\mathbf{x}), \boldsymbol{\Sigma}_{\phi}(\mathbf{x}))$ and $p_{\theta}(\mathbf{x} | \mathbf{z}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_{\theta}(\mathbf{z}), \sigma^2 I_d)$, the reconstruction term simplifies to:

$$\mathbb{E}_{\mathbf{Z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x} | \mathbf{Z})] = -\frac{1}{2\sigma^2} \mathbb{E}_{\mathbf{Z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} [\|\mathbf{x} - \boldsymbol{\mu}_{\theta}(\mathbf{Z})\|^2] - \frac{d}{2} \log(2\pi\sigma^2). \quad (26)$$

- ▶ From this perspective, the second term, *KL-penalty term*, acts as a regularizer in the training of autoencoders.

Variational Autoencoder

- ▶ The first term can be approximated via the Monte Carlo method:

$$\int (\log p_{\theta}(\mathbf{x}|\mathbf{z})) q_{\phi}(\mathbf{z}|\mathbf{x}) d\mathbf{z} \approx M^{-1} \sum_{m=1}^M \log p_{\theta}(\mathbf{x}|\mathbf{z}^{(m)}) \quad (27)$$

where $\mathbf{z}^{(m)} \stackrel{\text{i.i.d.}}{\sim} q_{\phi}(\mathbf{z}|\mathbf{x})$, and

$$\log p_{\theta}(\mathbf{x}|\mathbf{z}^{(m)}) = -\frac{1}{2\sigma^2} \sum_{j=1}^d (x_j - (\mu_{\theta}(\mathbf{z}_m))_j)^2 - \frac{d}{2} \log(2\pi\sigma^2). \quad (28)$$

- ▶ Under some conditions, this formulation reduces to the reconstruction error of vanilla autoencoders.
- ▶ In sampling $\mathbf{z}^{(m)} \sim q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_{\phi}(\mathbf{x}), \Sigma_{\phi}(\mathbf{x}))$, *reparameterization trick* is applied: $\mathbf{Z}^{(m)} = \boldsymbol{\mu}_{\phi}(\mathbf{x}) + \Sigma_{\phi}(\mathbf{x})^{1/2} \boldsymbol{\epsilon}^{(m)}$ where $\boldsymbol{\epsilon}^{(m)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$.

Variational Autoencoder

- ▶ For the second term, the KL penalty term $\mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$, we can use the following formula:

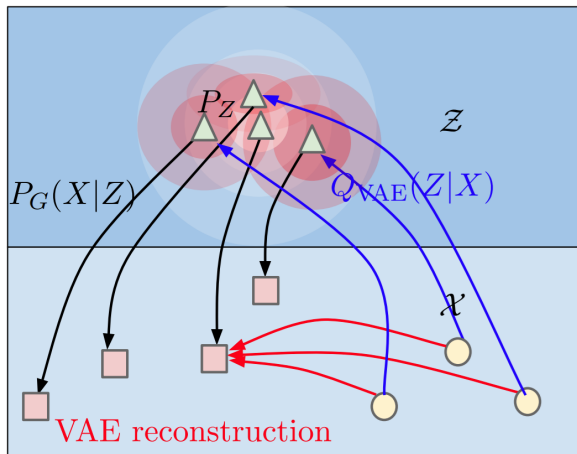
$$\begin{aligned} & \mathcal{D}_{\text{KL}}(\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_1, \Sigma_1) || \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_2, \Sigma_2)) \\ &= \frac{1}{2} \left((\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T \Sigma_2^{-1} (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) + \text{tr}(\Sigma_2^{-1} \Sigma_1) - \log \frac{|\Sigma_1|}{|\Sigma_2|} - d \right). \end{aligned} \quad (29)$$

See StatProofBook contributors (2026) for details on the proof.

- ▶ When $p(\mathbf{z}) = \mathcal{N}(0, I_d)$ and $\Sigma_{\phi}(\mathbf{x}) = \text{diag}(\sigma_{\phi}^2(\mathbf{x}))$, the KL penalty term simplifies to:

$$\mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z} | \mathbf{x}) || p(\mathbf{z})) = \frac{1}{2} \sum_{j=1}^d \left((\mu_{\phi}(\mathbf{x}))_j^2 + (\sigma_{\phi}^2(\mathbf{x}))_j - \log(\sigma_{\phi}^2(\mathbf{x}))_j - 1 \right). \quad (30)$$

Variational Autoencoder



The figure is from Tolstikhin et al. (2017).

Variational Autoencoder

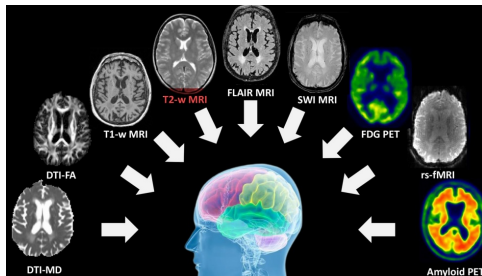


Images are edited from Kingma and Welling (2014).

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Motivation



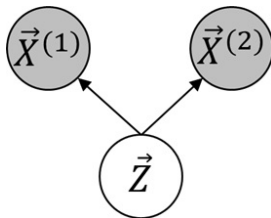
- ▶ Data often have multi-modal structures.
- ▶ Multi-modalities share comprehensive information while each modality also possesses its unique information.

The figure is from Gong et al. (2023)

Linear Method

- ▶ (Traditional) Linear methods are usually based on low-rank matrix compression theory:
 1. Multi-set Canonical Correlation Analysis (Kettenring, 1971)
 2. Joint and Individual Variation Explained (Lock et al., 2013)
 3. Multi-block Partial Least Squares (Chen et al., 2009).
- ▶ Multi-modal data may have complicated nonlinear dependencies given latent factors, and we may need to learn mixing mechanisms to analyze new test data.

Joint Multi-modal VAE

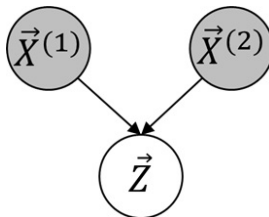


- ▶ Joint Multi-modal VAEs (Suzuki et al., 2016) are one of the earliest VAEs-based method, which learn the structure of bi-modal data.
- ▶ They assume that two modalities are generated from the same representations:

$$p_{\theta}(\mathbf{z}, \mathbf{x}^{(1)}, \mathbf{x}^{(2)}) := p(\mathbf{z})p_{\theta}(\mathbf{x}^{(1)}|\mathbf{z})p_{\theta}(\mathbf{x}^{(2)}|\mathbf{z}) \quad (31)$$

where $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ are realizations of the bi-modal data.

Joint Multi-modal VAE



- For the inference, they introduce a joint encoder to approximate the posterior distribution of latent variables given the bi-modal data:

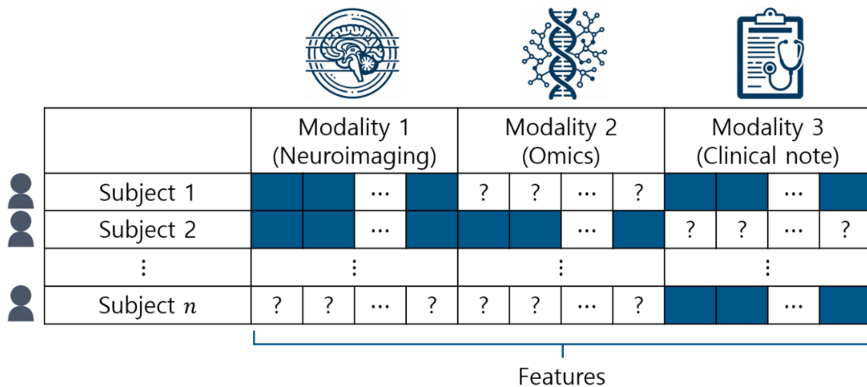
$$q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)}). \quad (32)$$

Joint Multi-modal VAE

- ▶ Given the generation and inference processes, the loss function, the negative ELBO w.r.t. $\log p_{\theta}(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})$ can be expressed as:

$$\begin{aligned}
 & -\log p_{\theta}(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) + \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)})||p_{\theta}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)})) \\
 &= -\int (\log p_{\theta}(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}|\mathbf{z})) q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) d\mathbf{z} + \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)})||p(\mathbf{z})) \\
 &= -\int (\log p_{\theta}(\mathbf{x}^{(1)}|\mathbf{z}) + \log p_{\theta}(\mathbf{x}^{(2)}|\mathbf{z})) q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) d\mathbf{z} \\
 &\quad + \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)})||p(\mathbf{z})).
 \end{aligned}
 \tag{33}$$

Joint Multi-modal VAE



- ▶ *Block-missing*, a situation where some modalities are absent, is a chronic problem in multi-modal data analysis (Song et al., 2024).
- ▶ Joint Multi-modal VAEs introduce modality-specific encoders to alleviate this issue.

Joint Multi-modal VAE

- ▶ Let $q_{\phi^{(1)}}(\mathbf{z}|\mathbf{x}^{(1)})$ and $q_{\phi^{(2)}}(\mathbf{z}|\mathbf{x}^{(2)})$ be modality-specific encoders.
- ▶ The modified objective function can be expressed as:

$$\begin{aligned}
 & - \int (\log p_{\theta}(\mathbf{x}^{(1)}|\mathbf{z}) + \log p_{\theta}(\mathbf{x}^{(2)}|\mathbf{z})) q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) d\mathbf{z} + \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) || p(\mathbf{z})) \\
 & + \alpha (\mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) || q_{\phi^{(1)}}(\mathbf{z}|\mathbf{x}^{(1)})) + \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) || q_{\phi^{(2)}}(\mathbf{z}|\mathbf{x}^{(2)}))).
 \end{aligned}
 \tag{34}$$

- ▶ Note that all the posterior distributions, $q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \mathbf{x}^{(2)})$, $q_{\phi^{(1)}}(\mathbf{z}|\mathbf{x}^{(1)})$, and $q_{\phi^{(2)}}(\mathbf{z}|\mathbf{x}^{(2)})$, are modeled as Gaussians, the new added terms have closed-form expressions.

Joint Multi-modal VAE

Input	Generated attributes
	Male : 0.95 Eyeglasses : -0.99 Young : 0.30 Smiling : -0.97 . .
	Male : 0.22 Eyeglasses : -0.99 Young : 0.87 Smiling : -1.00 . .

- Imputation results using $q_{\phi(1)}(\mathbf{z}|\mathbf{x}^{(1)})$ and $p_{\theta}(\mathbf{x}^{(2)}|\mathbf{z})$.

The figure is from Suzuki et al. (2016).

From Bi-modal to Multi-modal

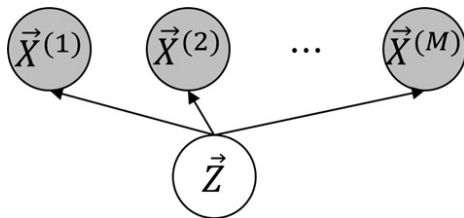
- What happens when the number of modality M increases? When we directly extend the loss function of joint multi-modal VAEs, we have:

$$\begin{aligned}
 & -\log p_{\theta}(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(M)}) + \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(M)})||p_{\theta}(\mathbf{z}|\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(M)})) \\
 &= -\int \left(\sum_{m=1}^M \log p_{\theta}(\mathbf{x}^{(m)}|\mathbf{z}) \right) q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(M)}) d\mathbf{z} + \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(M)})||p(\mathbf{z})).
 \end{aligned}
 \tag{35}$$

Q: What would be the joint encoder network size for this approach?

Q: Let's consider adding encoders for each observed modality pair. What is the number of possible missing types?

Multi-modal VAE

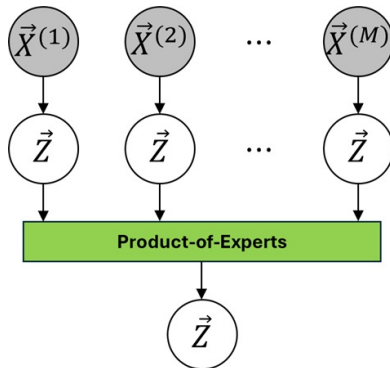


- ▶ Multi-modal VAEs (Wu and Goodman, 2018) introduce modality-specific VAEs and aggregate information from each modality using product-of-experts techniques (Hinton, 2002).
- ▶ Their generation process can be expressed as:

$$p_{\theta^{(1:M)}}(\mathbf{z}, \mathbf{x}^{(1:M)}) := p(\mathbf{z}) \prod_{m=1}^M p_{\theta^{(m)}}(\mathbf{x}^{(m)}|\mathbf{z}) \quad (36)$$

where $\theta^{(1:M)} := (\theta^{(1)}, \dots, \theta^{(M)})^T$ and $\mathbf{x}^{(1:M)} := (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(M)})^T$.

Multi-modal VAE



- ▶ Product-of-experts techniques enable us to use modality-specific inference networks to build an inference model incorporating all the observed modalities.

Multi-modal VAE

- ▶ When all modalities are observed, the (variational) inference aims to approximate:

$$p_{\theta(1:M)}(\mathbf{z}|\mathbf{x}^{(1:M)}) \propto \frac{\prod_{m=1}^M p_{\theta(m)}(\mathbf{z}|\mathbf{x}^{(m)})}{p(\mathbf{z})^{M-1}} \quad (37)$$

Multi-modal VAE

- ▶ When each modality-specific encoder perfectly approximates the corresponding posterior, i.e., $q_{\phi(m)}(\mathbf{z}|\mathbf{x}^{(m)}) \approx p_{\theta(m)}(\mathbf{z}|\mathbf{x}^{(m)})$ for all m , we can utilize them to approximate $p_{\theta(1:M)}(\mathbf{z}|\mathbf{x}^{(1:M)})$ using Equation (37).
- ▶ In implementation, multi-modal VAEs model the quotient $\tilde{q}_{\phi(m)}(\mathbf{z}|\mathbf{x}^{(m)}) := q_{\phi(m)}(\mathbf{z}|\mathbf{x}^{(m)})/p(\mathbf{z})$ instead of $q_{\phi(m)}(\mathbf{z}|\mathbf{x}^{(m)})$.
- ▶ With quotient networks, the RHS in Equation (37) can be expressed as $p(\mathbf{z}) \prod_{m=1}^M \tilde{q}_{\phi(m)}(\mathbf{z}|\mathbf{x}^{(m)})$. All the distributions of prior and quotient networks are Gaussian, implying a closed-expression.

Multi-modal VAE: Loss Function

► The loss function consists of three negative ELBOs.

1. $(\mathbf{X}^{(1:M)})$ The negative ELBO w.r.t. $\log p_{\theta(1:M)}(\mathbf{x}^{(1:M)})$:

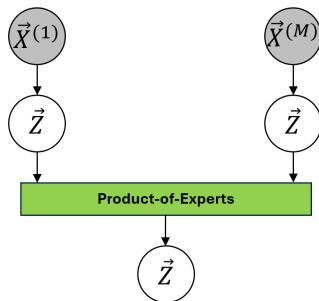
$$\begin{aligned}
 & -\log p_{\theta(1:M)}(\mathbf{x}^{(1:M)}) + \mathcal{D}_{\text{KL}}(q_{\phi(1:M)}(\mathbf{z}|\mathbf{x}^{(1:M)})||p_{\theta(1:M)}(\mathbf{z}|\mathbf{x}^{(1:M)})) \\
 &= -\int \left(\sum_{m=1}^M \log p_{\theta(m)}(\mathbf{x}^{(m)}|\mathbf{z}) \right) q_{\phi(1:M)}(\mathbf{z}|\mathbf{x}^{(1:M)}) d\mathbf{z} + \mathcal{D}_{\text{KL}}(q_{\phi(1:M)}(\mathbf{z}|\mathbf{x}^{(1:M)})||p(\mathbf{z}))
 \end{aligned} \tag{38}$$

where $q_{\phi(1:M)}(\mathbf{z}|\mathbf{x}^{(1:M)}) \propto p(\mathbf{z}) \prod_{m=1}^M \tilde{q}_{\phi(m)}(\mathbf{z}|\mathbf{x}^{(m)})$.

2. $(\mathbf{X}^{(m)})$ The negative ELBO w.r.t. $\log p_{\theta(m)}(\mathbf{x}^{(m)})$

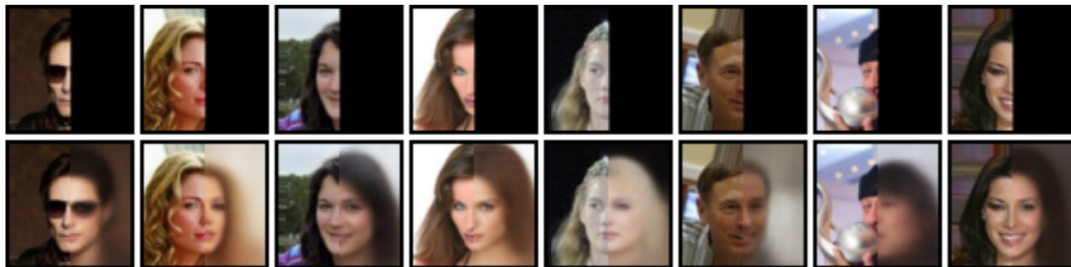
3. $(\mathbf{X}^{(\pi(1):\pi(m))})$ The negative ELBO w.r.t. $\log p_{\theta(\pi(1):\pi(m))}(\mathbf{x}^{(\pi(1):\pi(m))})$ where $\pi(1), \dots, \pi(m)$ are the first m indices from randomly permuted indices.

Multi-modal VAE



- ▶ For example, for a sample having only $\mathbf{X}^{(1)}$ and $\mathbf{X}^{(3)}$, the loss is the summation of the joint ELBO with respect to $(\mathbf{x}^{(1)}, \mathbf{x}^{(3)})^T$ and the marginal ELBOs with respect to $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(3)}$.

Multi-modal VAE



The figure is from Wu and Goodman (2018).

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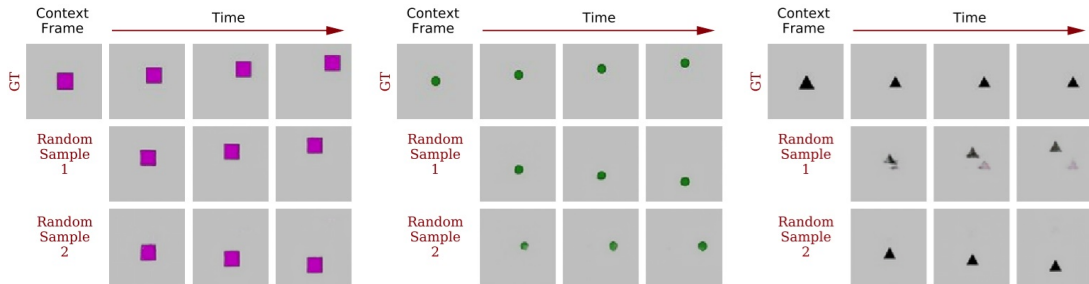
Motivation



- Video frames are popular temporal data, i.e., a time series of image frames at each time step.

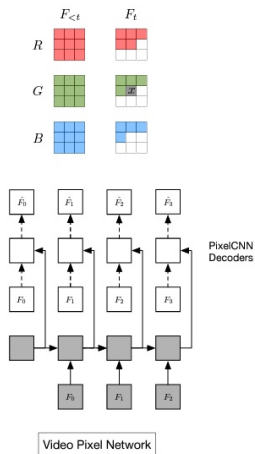
Examples in video generation. Images are from Szeto et al. (2019).

Motivation



Images are edited from Babaeizadeh et al. (2017).

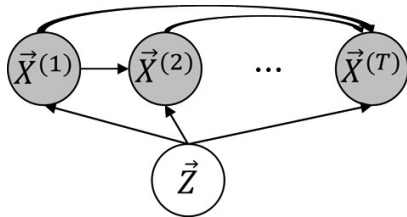
Auto-Regressive Model



- ▶ Lotter et al. (2016) proposed a deterministic method that minimizes the MSE between true and predicted future frames.
- ▶ This approach can be viewed as an auto-regressive model.
- ▶ Video Pixel Network (Kalchbrenner et al., 2017) extended image auto-regressive models to videos.

The figure is from Kalchbrenner et al. (2017).

Stochastic Variational Video Prediction

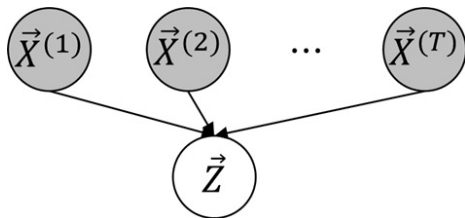


- ▶ Stochastic Variational Video Prediction (SV2P) (Babaeizadeh et al., 2017) is one of the earliest VAE-based video generation methods. It proposes two latent variable models: using (i) time-invariant $\mathbf{Z}$ and (ii) time-variant $\mathbf{Z}^{(t)}$.
- ▶ For (i), the generative model can be expressed as:³

$$p_{\theta}(\mathbf{z}, \mathbf{x}^{(1:T)}) = p(\mathbf{z})p_{\theta}(\mathbf{x}^{(1)}|\mathbf{z})p_{\theta}(\mathbf{x}^{(2)}|\mathbf{z}, \mathbf{x}^{(1)}) \cdots p_{\theta}(\mathbf{x}^{(T)}|\mathbf{z}, \mathbf{x}^{(1:(T-1))}). \quad (39)$$

³The SV2P assumes that first few image frames are available at test time.

Stochastic Variational Video Prediction

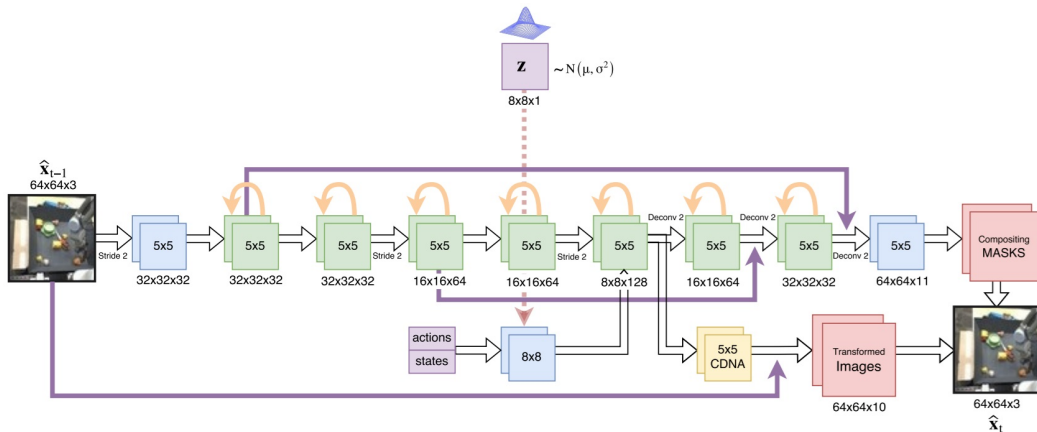


- SV2P uses $q_\phi(\mathbf{z}|\mathbf{x}^{(1:T)})$ to minimize the negative ELBO w.r.t. $\log p_\theta(\mathbf{x}^{(1:T)})$:

$$\begin{aligned}
 & -\log p_\theta(\mathbf{x}^{(1:T)}) + \mathcal{D}_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}^{(1:T)})||p_\theta(\mathbf{z}|\mathbf{x}^{(1:T)})) \\
 &= -\int \left(\log p_\theta(\mathbf{x}^{(1:T)}|\mathbf{z}) \right) q_\phi(\mathbf{z}|\mathbf{x}^{(1:T)}) d\mathbf{z} + \mathcal{D}_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}^{(1:T)})||p(\mathbf{z})) \\
 &= -\sum_{t=1}^T \int \left(\log p_\theta(\mathbf{x}^{(t)}|\mathbf{z}, \mathbf{x}^{(1:(t-1))}) \right) q_\phi(\mathbf{z}|\mathbf{x}^{(1:T)}) d\mathbf{z} + \mathcal{D}_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}^{(1:T)})||p(\mathbf{z})).
 \end{aligned}$$

(40)

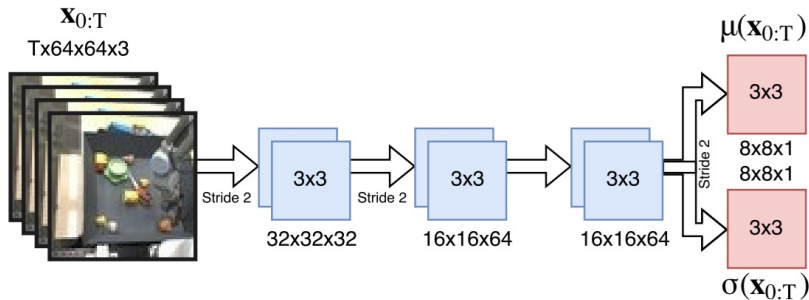
Stochastic Variational Video Prediction



- $p_{\theta}(\mathbf{x}^{(t)} | \mathbf{z}, \mathbf{x}^{(1:(t-1))})$: The parameter θ is shared across all time steps t .

The figure is edited from Babaeizadeh et al. (2017).

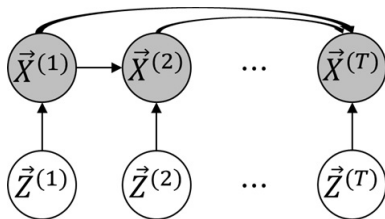
Stochastic Variational Video Prediction



- $q_{\phi}(\mathbf{z}|\mathbf{x}^{(1:T)})$: The inference network uses all the image frames to infer the time-invariant latent variable.

The figure is from Babaeizadeh et al. (2017).

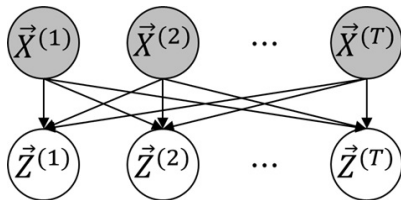
Stochastic Variational Video Prediction



- For the case of (ii) a time-variant latent variable $\mathbf{Z}^{(t)}$, the SV2P model formulates the joint distribution as follows:

$$p_{\theta}(\mathbf{z}^{(1:T)}, \mathbf{x}^{(1:T)}) = \prod_{t=1}^T p(\mathbf{z}^{(t)}) \prod_{t=1}^T p_{\theta}(\mathbf{x}^{(t)} | \mathbf{z}^{(t)}, \mathbf{x}^{(1:(t-1))}). \quad (41)$$

Stochastic Variational Video Prediction

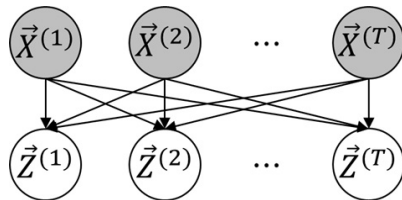


- ▶ The time-variant $\mathbf{Z}^{(t)}$ is inferred using the full sequence of frames $\mathbf{X}^{(1:T)}$.
- ▶ The inference network can be formalized as

$$q_{\phi}(\mathbf{z}^{(1:T)} \mid \mathbf{x}^{(1:T)}) = \prod_{t=1}^T q_{\phi}(\mathbf{z}^{(t)} \mid \mathbf{x}^{(1:T)}). \quad (42)$$

- ▶ Here, the conditional functional form $q_{\phi}(\cdot \mid \mathbf{x}^{(1:T)})$ is assumed to be time-invariant. That is, conditioned on the observed video $\mathbf{X}^{(1:T)}$, the latent variables $\mathbf{Z}^{(1)}, \dots, \mathbf{Z}^{(T)}$ are conditionally i.i.d.

Stochastic Variational Video Prediction

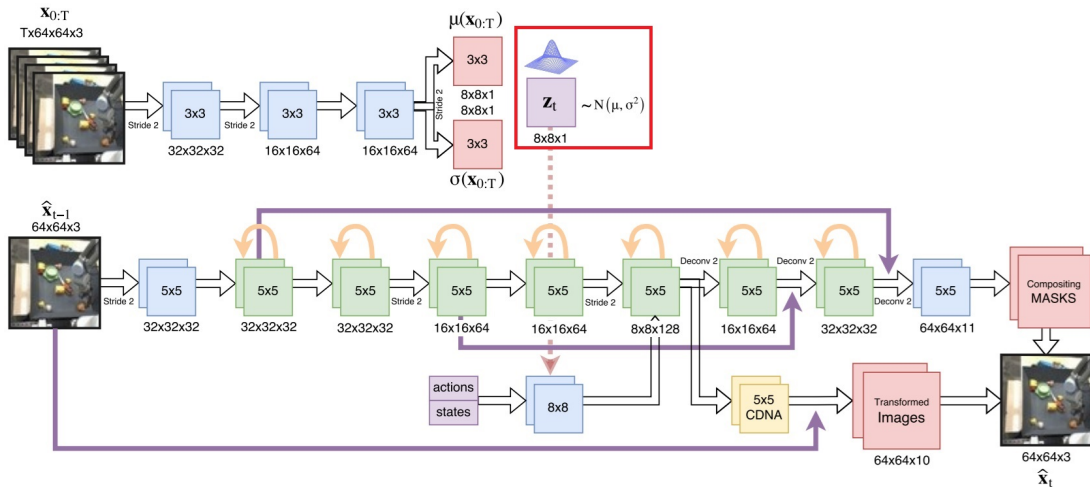


- The loss function is the negative ELBO w.r.t. $\log p_{\theta}(\mathbf{x}^{(1:T)})$:

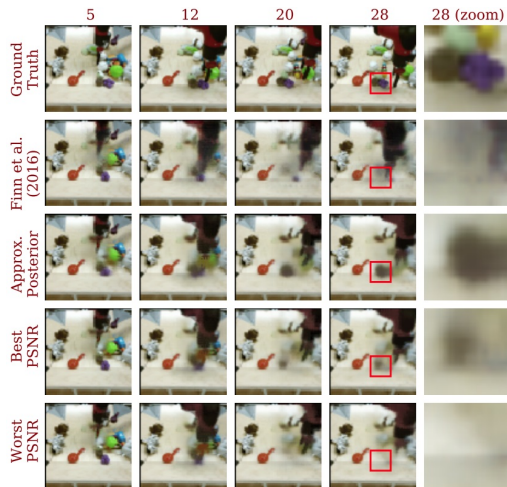
$$\begin{aligned}
 & -\log p_{\theta}(\mathbf{x}^{(1:T)}) + \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}^{(1:T)}|\mathbf{x}^{(1:T)})||p_{\theta}(\mathbf{z}^{(1:T)}|\mathbf{x}^{(1:T)})) \\
 &= \sum_{t=1}^T \left(-\int (\log p_{\theta}(\mathbf{x}^{(t)}|\mathbf{z}^{(t)}, \mathbf{x}^{(1:(t-1))})) q_{\phi}(\mathbf{z}^{(t)}|\mathbf{x}^{(1:T)}) d\mathbf{z}^{(t)} \right. \\
 & \quad \left. + \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z}^{(t)}|\mathbf{x}^{(1:T)})||p(\mathbf{z}^{(t)})) \right)
 \end{aligned} \tag{43}$$

where $q_{\phi}(\mathbf{z}^{(1:T)}|\mathbf{x}^{(1:T)}) = \prod_{t=1}^T q_{\phi}(\mathbf{z}^{(t)}|\mathbf{x}^{(1:T)})$.

Stochastic Variational Video Prediction

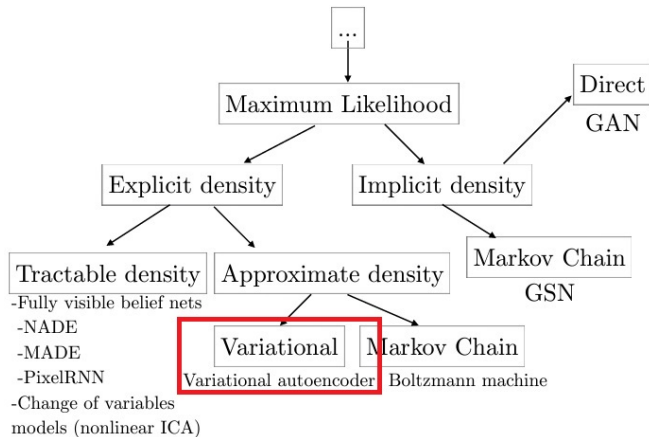


Stochastic Variational Video Prediction



Images are edited from Babaeizadeh et al. (2017). PSNR stands for Peak Signal-to-Noise Ratio.

Lecture Topics



The figure is edited from Goodfellow (2016).

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